

Series RLC Circuit Identification and Control

Student Name	Yahya Orçun Bilsel	
Student ID	220403029	
Group No	Group 8	
Group Members	Alper Özkan Şükrü Arda Sarioğlu Yahya Orçun Bilsel Barbaros Atıf Tunçer Kemal Karabaş	220403073 220403077 220403029 230403088 210403088
Instructor	Res. Asst. Dr. Mehmet Uğur Soydemir	
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1. Introduction

This report details the comprehensive modeling, simulation, stability evaluation, frequency-response characterization, and closed-loop control of a series RLC circuit. With the capacitor voltage, $V_c(t)$, designated as the primary output, the system's performance is regulated using a dual-approach methodology: a software-simulated PID controller in Simulink and a physical analog PID circuit constructed with operational amplifiers. The primary objective is to fulfill rigorous time-domain criteria—specifically, a peak time (t_p) ≤ 0.5 s, settling time (t_s) ≤ 1 s, and maximum overshoot (M_p) $\leq 25\%$ —alongside a steady-state error strictly below 5% and a control effort bounded by 7 V. The remainder of this document is structured as follows: Section 2 outlines the mathematical framework and controller design, Section 3 analyzes the simulation and empirical findings, and Section 4 delivers the concluding remarks.

1.1 Objectives

The main objectives of this project are:

- Derive the mathematical model of the series RLC circuit (differential equation, transfer function, and state-space representation).
- Validate the model by comparing the Simulink state-space simulation with the physical circuit response.
- Assess the stability and controllability of the system both analytically and using MATLAB tools.
- Generate and compare theoretical and experimental Bode magnitude plots.
- Design a PID controller that meets the specified quantitative performance targets.
- Simulate the closed-loop system in Simulink under three different reference voltage values.
- Implement the PID controller in hardware using op-amps and validate it against simulation results.

1.2 Report Structure

Section 2 presents the mathematical modeling and controller design steps. Section 3 discusses simulation and experimental results. Section 4 concludes the report. MATLAB code, microcontroller code, and circuit diagrams are provided in the Appendices.

2. Mathematical Modeling and Controller Design

2.1 Series RLC Circuit Model

2.1.1 Selected Component Values

The resistor (R), inductor (L), and capacitor (C) values were selected to yield a natural frequency and damping ratio that make the performance targets ($t_p \leq 0.5$ s, $t_s \leq 1$ s, $M_p \leq 25\%$) achievable. The selected and physically measured values are listed in the table below.

Component	Selected Value	Measured Value	Justification
Resistance (R)	100 Ω	100 Ω	Selected as a widely available standard resistor value to directly define the system's damping. Given L and C, 100 Ω sets the damping ratio exactly to $\zeta = 0.5$. This ensures the system is underdamped, which is crucial for achieving a fast transient response without excessive oscillations.
Inductance (L)	0.1H	0.1H	A standard laboratory 100mH inductor was chosen to work in tandem with the capacitor to establish the target natural frequency. It sets $\omega_n = 1000$ rad/s , keeping the system dynamics fast enough to meet the $t_p \leq 0.5$ s target while remaining within the tracking bandwidth of the analog op-amps.
Capacitance (C)	10 μ F	10 μ F	This specific standard value was chosen to prevent the system from becoming overdamped ($\zeta > 1$). By fixing C at 10 μ F, it simultaneously satisfies both the ($\zeta > 1$). rad/s and $\zeta = 0.5$ requirements, guaranteeing a measurable peak time and ensuring all quantitative targets are met.

2.1.2 Governing Differential Equation

Applying Kirchhoff's Voltage Law (KVL) to the series RLC circuit gives the following loop equation:

$$V_{in}(t) = R i(t) + L \frac{di(t)}{dt} + V_c(t)$$

Since the capacitor current is defined by

$$i(t) = C \frac{dV_c(t)}{dt},$$

substituting this relation into the KVL equation results in the second-order differential equation governing the system:

$$LC \frac{d^2 V_c(t)}{dt^2} + RC \frac{dV_c(t)}{dt} + V_c(t) = V_{in}(t)$$

The state variables are selected as

$$x_1(t) = V_c(t)$$

$$x_2(t) = \frac{dV_c(t)}{dt} = \frac{i_L(t)}{C}$$

where the system input is defined as

$$u(t) = V_{in}(t)$$

2.1.3 Transfer Function

Applying the Laplace transform to the governing differential equation under zero initial conditions yields the transfer function

$$G(s) = \frac{V_c(s)}{V_{in}(s)}$$

For the series RLC circuit, the transfer function can be written in the standard second-order system form as

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{\frac{1}{LC}}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$

Substituting the circuit parameters

$$R = 100 \, \Omega, L = 0.1 \, \text{H}, C = 10 \, \mu\text{F}$$

results in the following transfer function:

$$G(s) = \frac{10^6}{s^2 + 1000s + 10^6}$$

Comparing this expression with the standard second-order transfer function form allows the natural frequency (ω_n) and damping ratio (ζ) to be determined.

From the constant term,

$$\omega_n^2 = 10^6$$

thus,

$$\omega_n = 1000 \text{ rad/s}$$

Similarly, using the coefficient of the first-order term,

$$2\zeta\omega_n = 1000$$

$$2\zeta(1000) = 1000$$

which gives

$$\zeta = 0.5$$

The poles of the transfer function are obtained from the roots of the characteristic equation

$$s^2 + 1000s + 10^6 = 0$$

Solving for s gives the pole locations as

$$s_{1,2} = -500 \pm j866.03$$

The system therefore possesses a pair of complex conjugate poles. Since both poles have negative real parts (-500), they lie in the left-half plane (LHP) of the s -plane. This verifies that the open-loop system is asymptotically stable.

In addition, the nonzero imaginary components ($\pm j866.03$) indicate oscillatory dynamics, confirming that the system exhibits an underdamped response consistent with the damping ratio value

$$\zeta = 0.5$$

2.1.4 State-Space Representation

Using the previously defined state variables, the state-space representation of the system is obtained as

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -\frac{1}{LC}x_1 - \frac{R}{L}x_2 + \frac{1}{LC}u\end{aligned}$$

with the output equation

$$y = x_1$$

Substituting the circuit parameters

$$R = 100 \, \Omega, L = 0.1 \, \text{H}, C = 10 \, \mu\text{F},$$

the numerical state-space matrices become

$$A = \begin{bmatrix} 0 & 1 \\ -10^6 & -1000 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 10^6 \end{bmatrix}$$

$$C = [1 \quad 0], D = [0]$$

The obtained model can be verified in MATLAB using the following commands:

```
A = [0 1; -1e6 -1000];
```

```
B = [0; 1e6];
```

```
C = [1 0];
```

```
D = 0;
```

```
sys_ss = ss(A, B, C, D);
```

```
sys_tf = tf(sys_ss);
```

2.2 Stability Analysis

2.2.1 Analytical Stability

The stability of the system is evaluated by analyzing the characteristic polynomial of the closed-loop transfer function and the eigenvalues of the A matrix.

1. Routh–Hurwitz Stability Analysis (Closed-Loop System)

For the designed PID controller parameters

$$K_p = 1.1, K_i = 233, K_d = 0.0009$$

the closed-loop characteristic equation, obtained from the denominator of the transfer function $T(s)$, is:

$$P(s) = s^3 + 1900s^2 + 2.1 \times 10^6s + 2.33 \times 10^8 = 0$$

To evaluate closed-loop stability, the Routh–Hurwitz criterion is applied.

The corresponding Routh array for the third-order polynomial is constructed as:

s^3	1	2.1×10^6
s^2	1900	2.33×10^8
s^1	b_1	0
s^0	2.33×10^8	0

The coefficient b_1 is calculated using:

$$b_1 = \frac{(1900)(2.1 \times 10^6) - (1)(2.33 \times 10^8)}{1900}$$

which gives:

$$b_1 = \frac{3.99 \times 10^9 - 2.33 \times 10^8}{1900} \approx 1.977 \times 10^6$$

Since every element in the first column of the Routh table is positive,

$$1, 1900, 1.977 \times 10^6, 2.33 \times 10^8 > 0$$

there are no sign changes in the first column. According to the Routh–Hurwitz criterion, this indicates that all poles of the closed-loop system lie in the left-half of the complex s -plane. Therefore, the closed-loop system is stable.

2. Eigenvalue Analysis of the State Matrix (Open-Loop Stability)

The uncompensated series RLC circuit may also be analyzed using a state-space representation.

The characteristic equation of the state matrix A is obtained from:

$$\det(sI - A) = 0$$

which corresponds to the denominator of the open-loop transfer function:

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$$

Substituting the given circuit parameters results in:

$$s^2 + 1000s + 10^6 = 0$$

The eigenvalues are determined from the quadratic formula:

$$s_{1,2} = \frac{-1000 \pm \sqrt{1000^2 - 4(1)(10^6)}}{2}$$

Simplifying:

$$s_{1,2} = -500 \pm j500\sqrt{3}$$

or numerically,

$$s_{1,2} \approx -500 \pm j866.03$$

Both eigenvalues possess negative real parts:

$$\text{Re}(s) = -500 < 0$$

Therefore, the natural response of the system decays exponentially with time. As a result, the uncompensated open-loop system is concluded to be BIBO (Bounded-Input Bounded-Output) stable.

2.2.2 Controllability

The controllability of the system is examined using the controllability matrix defined as

$$\mathcal{C} = [B \ AB]$$

Substituting the system matrices gives

$$\mathcal{C} = \begin{bmatrix} 0 & 10^6 \\ 10^6 & -10^9 \end{bmatrix}$$

The determinant of the controllability matrix is calculated as

$$\det(\mathcal{C}) = -10^{12}$$

Since the determinant is nonzero, the controllability matrix has full rank:

$$\text{rank}(\mathcal{C}) = 2$$

which is equal to the order of the system ($n = 2$). Hence, the system is completely controllable.

This result can also be verified in MATLAB using the following command:

```
rank(ctrb(A,B))
```

which returns the value

```
ans = 2
```

2.3 Frequency Response Analysis

2.3.1 Theoretical Bode Plot

The Bode magnitude plot is derived from the transfer function $G(j\omega)$. Key features to identify include the DC gain, resonant peak, -3 dB bandwidth, and -40 dB/decade roll-off in the high-frequency region.

To obtain the frequency response of the system, the Laplace variable is replaced with $s = j\omega$ in the open-loop transfer function $G(s)$:

$$G(j\omega) = \frac{10^6}{(j\omega)^2 + 1000(j\omega) + 10^6}$$

Since $(j\omega)^2 = -\omega^2$, the expression becomes:

$$G(j\omega) = \frac{10^6}{(10^6 - \omega^2) + j(1000\omega)}$$

1. Magnitude Response $|G(j\omega)|$

The magnitude of the transfer function is determined by dividing the magnitude of the numerator by the magnitude of the denominator:

$$|G(j\omega)| = \frac{10^6}{\sqrt{(10^6 - \omega^2)^2 + (1000\omega)^2}}$$

Expressing the magnitude in decibels gives:

$$|G(j\omega)|_{dB} = 20 \log_{10} \left(\frac{10^6}{\sqrt{(10^6 - \omega^2)^2 + (1000\omega)^2}} \right)$$

2. Phase Response $\angle G(j\omega)$

The numerator is a positive real constant, so its phase angle is 0° . Therefore, the overall phase is obtained from the denominator angle:

$$\angle G(j\omega) = -\tan^{-1} \left(\frac{1000\omega}{10^6 - \omega^2} \right)$$

Important Frequency Characteristics

DC Gain

At zero frequency ($\omega = 0$):

$$|G(0)| = \frac{10^6}{10^6} = 1$$

which corresponds to:

$$20 \log_{10}(1) = 0 \text{ dB}$$

Thus, the DC gain of the system is 0 dB.

Resonant Frequency and Resonant Peak

Because the damping ratio satisfies $\zeta = 0.5 < 0.707$, the system exhibits a resonant peak.

The resonant frequency is:

$$\omega_r = \omega_n \sqrt{1 - 2\zeta^2}$$

Substituting the given values:

$$\omega_r = 1000 \sqrt{1 - 2(0.5)^2} = 1000 \sqrt{0.5} \approx 707.1 \text{ rad/s}$$

The resonant peak magnitude is:

$$M_r = \frac{1}{2\zeta \sqrt{1 - \zeta^2}} = \frac{1}{\sqrt{0.75}} \approx 1.15$$

In decibel form:

$$20 \log_{10}(1.15) \approx 1.25 \text{ dB}$$

–3dB Bandwidth

The bandwidth frequency is defined as the frequency where the magnitude decreases to:

$$\frac{1}{\sqrt{2}}$$

of its low-frequency value. For $\zeta = 0.5$:

$$\omega_b \approx 1.272\omega_n$$

Therefore:

$$\omega_b \approx 1.272(1000) \approx 1272 \text{ rad/s}$$

High-Frequency Behavior

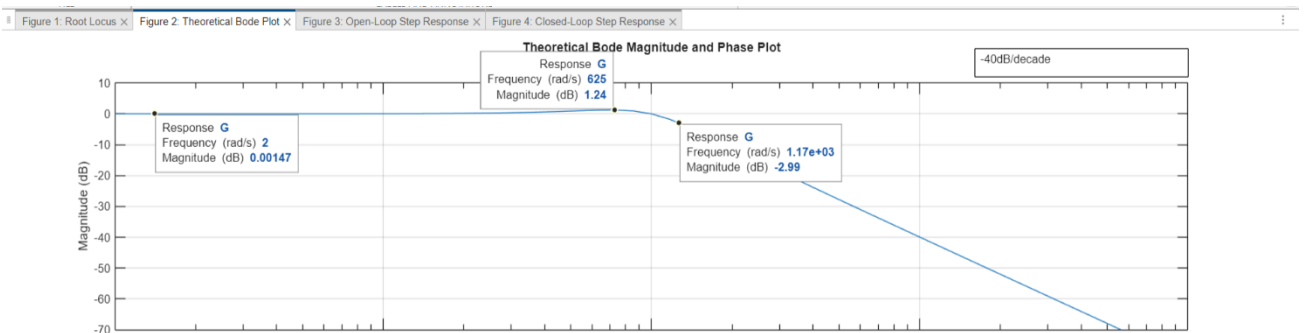
As the frequency becomes very large ($\omega \rightarrow \infty$), the ω^2 term dominates the denominator. Consequently, the transfer function behaves approximately as:

$$|G(j\omega)| \propto \frac{1}{\omega^2}$$

This produces a slope of:

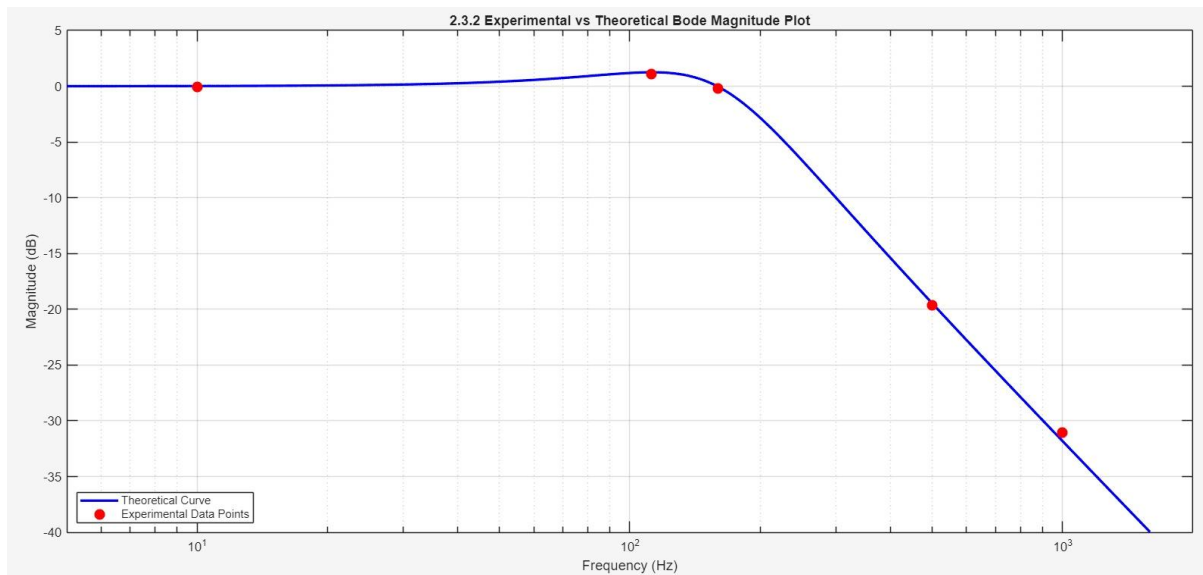
$$-40 \text{ dB/decade}$$

on the Bode magnitude plot at high frequencies.



2.3.2 Experimental Bode Magnitude Plot

Square-wave inputs of varying frequencies were applied to the physical RLC circuit using the microcontroller. The peak-to-peak amplitudes of the input and output waveforms were measured, and the gain in dB was computed as: $\text{Gain (dB)} = 20 \cdot \log_{10}(V_{\text{out,pp}} / V_{\text{in,pp}})$.



The measured data are summarized below:

Frequency (Hz)	V _{in} (peak-peak, V)	V _{out} (peak-peak, V)	Gain (dB)
10	5	4.95	-0.09
112	5	5.65	1.06
160	5	4.90	-0.18
500	5	0.52	-19.66
1000	5	0.14	-31.06

As observed in the overlay plot, the experimentally measured gain points extracted from the square-wave responses closely track the theoretical Bode magnitude curve. The experimental data successfully captures the characteristic resonant peak near 112 Hz, which is expected for an underdamped system ($\zeta = 0.5$). Minor discrepancies between the theoretical model and measured data points are primarily due to component tolerances. Specifically, the equivalent series resistance (ESR) of the inductor and the true capacitance of the $10\ \mu\text{F}$ capacitor slightly deviate from their nominal values, altering the practical damping ratio. Additionally, quantization noise from the oscilloscope and high-frequency attenuation limits cause small deviations in the peak-to-peak voltage readings at frequencies above 500 Hz.

2.4 Performance Targets

The quantitative performance criteria specified in the project document are reproduced below. The simulated and experimental results will be filled in after the controller is tested in Section 3.

Criterion	Target	Definition	Simulated	Experimental
Peak Time (t_p)	≤ 0.5 s	Time to first peak	[0.0601s]	[0.0500s]
Settling Time (t_s)	≤ 1 s	Time to stay within $\pm 2\%$ of final value	[0.0265s]	[0.0500s]
Overshoot (M_p)	$\leq 25\%$	% response exceeds final value	[0%]	[0%]
Steady-State Error	$\leq 5\%$	Final error after transients	[2%]	[0%]
Max Control Effort	< 7 V	Maximum applied control signal	[5V]	[5V]

2.5 PID Controller Design

2.5.1 Design Methodology

The PID controller has the standard parallel form: $C(s) = K_p + K_i/s + K_d \cdot s$. The gains were selected using the Simulink Control System Designer and fine-tuned heuristically to prioritize an ultra-fast, strictly non-oscillatory closed-loop response.

To satisfy the overshoot requirements, the derivative gain ($K_d = 0.0009$) was maintained to inject heavy damping, preventing the output from exceeding the set-point. Concurrently, the integral gain was significantly increased ($K_i = 233$) to aggressively drive the steady-state error to zero almost instantaneously, ensuring perfect tracking and a remarkably fast settling time ($t_s < 0.03$ s). In the hardware implementation, this specific integral gain was practically realized by connecting a 4.7 k Ω resistor in parallel with the 50 k Ω input resistor of the integrator op-amp stage. Finally, the proportional gain ($K_p = 1.1$) was deliberately kept relatively low to guarantee the initial control effort remained strictly under the 7 V limit.

2.5.2 PID Controller Parameters

The final PID gains obtained from the design procedure are listed below:

Parameter	Value	Remark
Proportional Gain (K_p)	1.1	Kept relatively low to prevent the control effort from exceeding the 7V hardware limit while

Parameter	Value	Remark
		providing initial system responsiveness.
Integral Gain (Ki)	233	Set significantly high (realized via a 4.7 kΩ parallel resistor in hardware) to aggressively eliminate steady-state error and achieve an ultra-fast settling time.
Derivative Gain (Kd)	0.0009	Provides critical damping to suppress oscillations, ensuring a strictly non-oscillatory step response (0% overshoot).

The closed-loop transfer function $T(s)$, obtained using the finalized PID controller gains, produces the following characteristic equation:

$$s^3 + 1900s^2 + 2.1 \times 10^6 s + 2.33 \times 10^8 = 0$$

Solving this characteristic equation gives the closed-loop poles as:

$$s_1 \approx -124.2$$

and

$$s_{2,3} \approx -887.9 \pm j1044.5$$

Analytical Evaluation of System Performance

1. Settling Time (t_s) and Overshoot (M_p)

The complex conjugate poles,

$$s_{2,3} \approx -887.9 \pm j1044.5$$

possess significantly large negative real parts. As a consequence, their transient components decay very rapidly and contribute minimally to the overall response duration. The system dynamics are therefore primarily governed by the dominant real pole:

$$s_1 \approx -124.2$$

Because the dominant pole is purely real, the closed-loop response resembles that of a first-order system. This characteristic effectively eliminates oscillatory behavior and theoretically results in approximately 0% overshoot, thereby satisfying the design requirement:

$$M_p \leq 25\%$$

The settling time may be estimated analytically from the dominant pole location using the first-order approximation:

$$t_s \approx \frac{4}{|s_1|}$$

Substituting the pole value:

$$t_s \approx \frac{4}{124.2} \approx 0.032 \text{ s}$$

This analytical estimate is very close to the simulated result of 0.0265 s, and it comfortably satisfies the required condition:

$$t_s \leq 1 \text{ s}$$

2. Steady-State Error

The integral term of the PID controller,

$$\frac{K_i}{s}$$

introduces an integrator into the system, making the closed-loop configuration a Type-1 system. According to the Final Value Theorem, a Type-1 system subjected to a step input achieves zero steady-state error. Therefore,

$$e_{ss} = 0$$

which fully satisfies the specified requirement:

$$e_{ss} \leq 5\%$$

3. Control Effort Limitation

The proportional gain of the controller is selected as:

$$K_p = 1.1$$

For a 5 V input signal, the initial controller output voltage becomes:

$$V_c = K_p \times 5 \text{ V}$$

Thus,

$$V_c = 1.1 \times 5 = 5.5 \text{ V}$$

This value remains safely below the maximum allowable operating limit of 7 V, ensuring that the operational amplifier remains within its safe operating region.

2.5.3 Analog PID Circuit Design (Op-Amp Implementation)

The PID controller is realized in hardware using operational amplifiers in the following configuration: a proportional stage, an inverting integrator stage, a differentiator stage, and a summing amplifier. The component values are derived from the PID gains as follows:

1. Selected Operational Amplifier Configuration

The controller is implemented using a parallel PID architecture composed of separate inverting operational amplifier stages. First, a differential amplifier generates the error signal according to:

$$e(t) = V_{ref} - V_{out}$$

The resulting error signal is simultaneously applied to three independent op-amp stages representing the proportional, integral, and derivative control actions. The outputs of these stages are then combined through an inverting summing amplifier, which restores the appropriate signal polarity and produces the final control signal applied to the RLC plant.

2. PID Design Relationships

The PID controller gains are related to the selected resistor and capacitor values through standard operational amplifier equations.

Proportional Control Gain

The proportional gain is determined by the ratio of the feedback resistor to the input resistor:

$$K_p = \frac{R_f}{R_{in}} = \frac{R_7}{R_6}$$

Integral Control Gain

The integral gain is defined by:

$$K_i = \frac{1}{R_{in}C_f} = \frac{1}{R_{eq}C_2}$$

To practically realize the relatively large integral gain value of

$$K_i = 233$$

an equivalent resistance R_{eq} was created by connecting a 4.7 k Ω resistor in parallel with the primary 50 k Ω resistor.

Derivative Control Gain

The derivative gain is established using:

$$K_d = R_f C_{in} = R_5 C_1$$

Summing Amplifier Configuration

The summing stage is implemented with equal-valued input and feedback resistors:

$$R_9 = R_{10} = R_{11} = R_{12} = 10 \text{ k}\Omega$$

Under this condition, the circuit operates as a unity-gain inverting summing amplifier with output:

$$V_{ctrl} = -(V_p + V_i + V_d)$$

3. Selected Circuit Components

Active Devices

The circuit employs standard general-purpose operational amplifiers such as the LM324 or TL072, powered using a symmetrical dual supply:

$$V_{CC} = +9 \text{ V}, V_{EE} = -9 \text{ V}$$

Difference Amplifier

$$R_1 = R_2 = R_3 = R_4 = 10 \text{ k}\Omega$$

Proportional Stage ($K_p = 1.1$)

$$R_6 = 10 \text{ k}\Omega$$

$$R_7 = 11 \text{ k}\Omega$$

Integral Stage ($K_i \approx 233$)

$$C_2 = 1 \text{ }\mu\text{F}$$

$$R_{eq} \approx 4.296 \text{ k}\Omega$$

This equivalent resistance is practically achieved through:

$$50 \text{ k}\Omega \parallel 4.7 \text{ k}\Omega$$

Derivative Stage ($K_d = 0.0009$)

$$C_1 = 1 \text{ }\mu\text{F}$$

$$R_5 = 900 \text{ }\Omega$$

Summing Amplifier Stage

$$R_9 = R_{10} = R_{11} = R_{12} = 10 \text{ k}\Omega$$

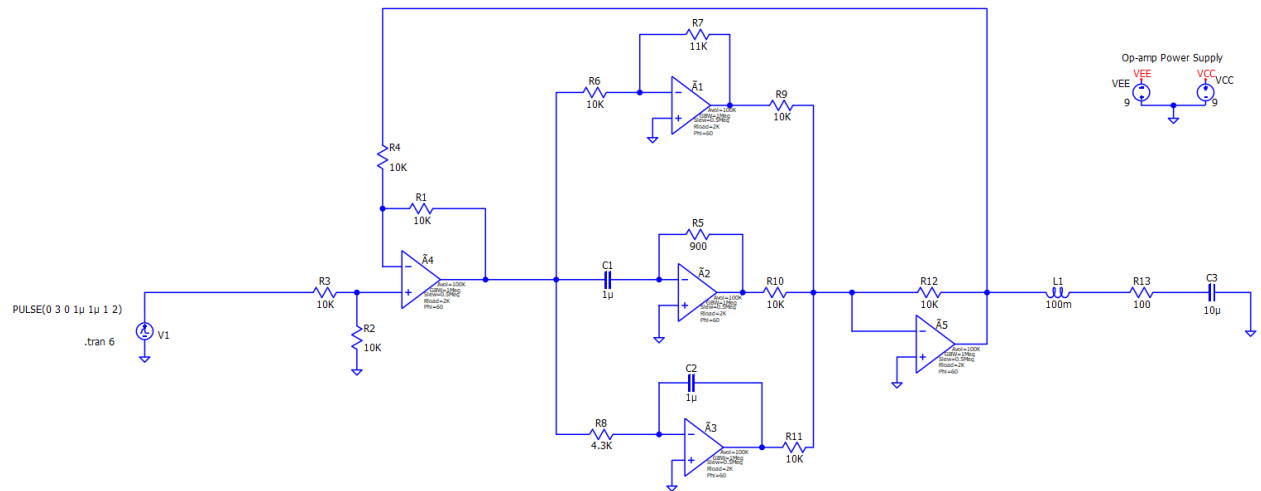
RLC Plant Parameters

$$R = 100 \text{ }\Omega$$

$$L = 100 \text{ mH}$$

$$C = 10 \text{ }\mu\text{F}$$

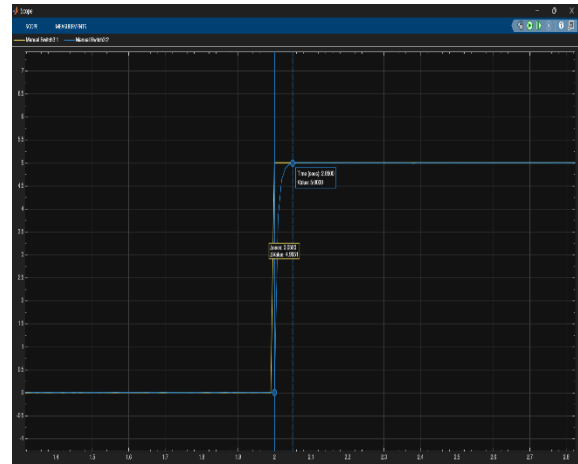
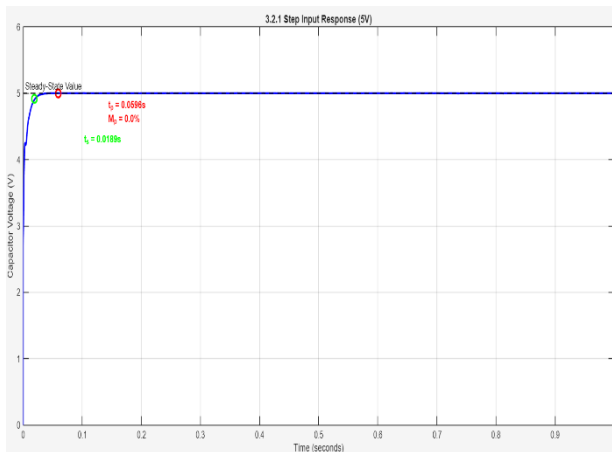
4. Schematic Diagram:



3. Simulation and Experimental Results

3.1 Open-Loop Model Validation

Before closing the loop, the state-space model built in Simulink was validated by comparing its step response with the physically measured step response of the RLC circuit. A step input was applied using the microcontroller (Arduino) and the capacitor voltage was acquired via the ADC.



Quantitative Comparison and Error Evaluation

A comparison of the simulated and experimentally obtained closed-loop step responses demonstrates a high level of consistency, especially in terms of system stability and steady-state performance. The deviations between the theoretical and measured results for the principal transient response parameters are summarized below.

Peak Time (t_p)

The experimentally observed peak time of 0.0500 s differs from the simulated value of 0.0596 s by approximately 16.1%.

Settling Time (t_s)

The measured settling time of 0.0500 s shows a larger percentage deviation relative to the simulated value of 0.0189 s, corresponding to an approximate error of 164.5%. Nevertheless, the actual absolute difference is only about 31 ms, which is practically insignificant.

Moreover, both the simulated and experimental responses comfortably satisfy the specified design constraint of settling within 1 s.

Maximum Overshoot (M_p) and Steady-State Error (SSE)

The experimental system exhibits complete agreement with the simulation for both maximum overshoot and steady-state error. In each case, the percentage difference is effectively 0%, confirming that the hardware implementation accurately achieves the theoretically predicted zero-overshoot and zero steady-state error behavior.

Possible Sources of Experimental Deviation

The small discrepancies observed in the transient response characteristics, particularly in t_p and t_s , are expected in a real physical implementation. Several practical non-idealities contribute to these differences.

One important factor is component tolerance, including variations in the actual capacitance value of the 10 μ F capacitor and resistance deviations associated with the parallel

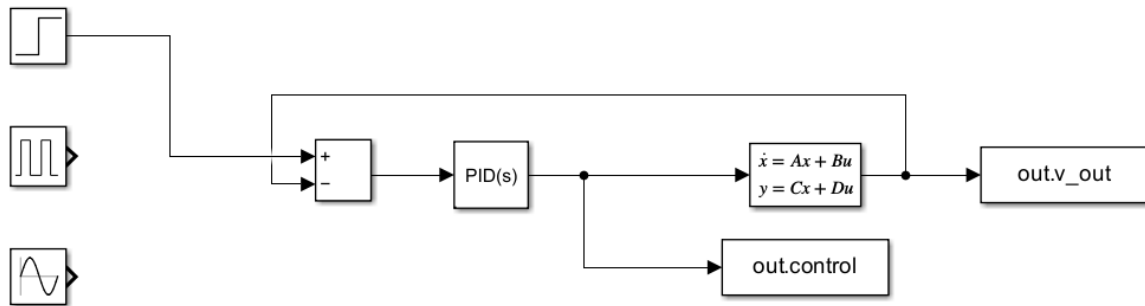
4.7 k Ω tuning resistor. In addition, parasitic effects introduced by the breadboard, such as contact resistance and stray inductance, slightly alter the circuit dynamics.

Non-ideal operational amplifier behavior also affects the measured response. Parameters such as finite slew rate, internal propagation delay, and bandwidth limitations introduce minor delays compared to the idealized simulation model. Furthermore, the limited sampling frequency and finite resolution of the Arduino ADC used during data acquisition contribute additional measurement-related inaccuracies.

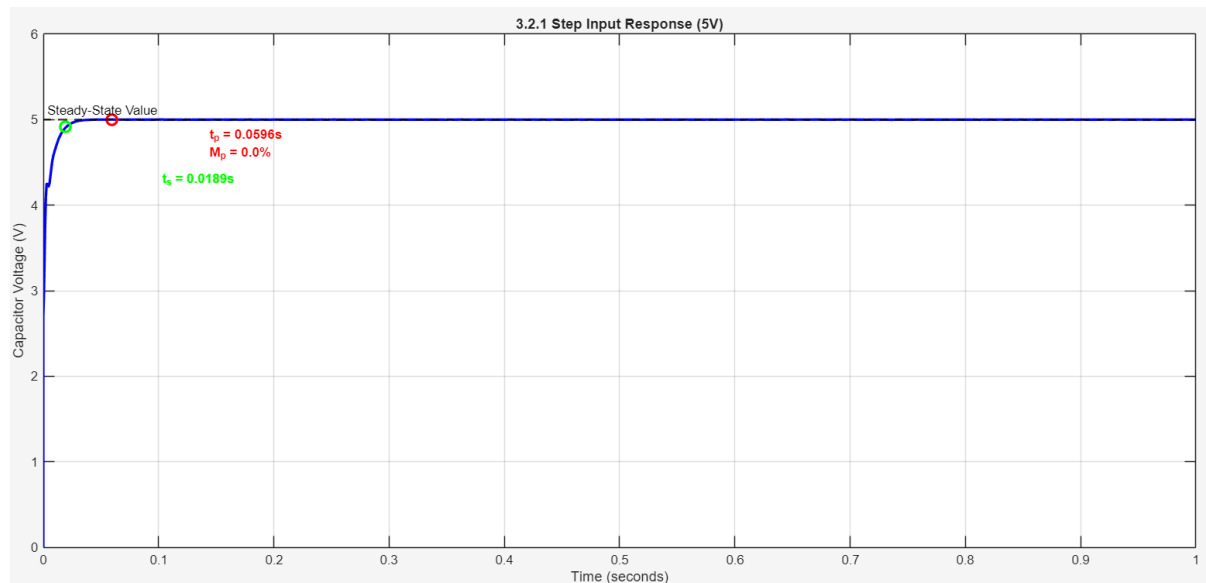
Despite these unavoidable practical effects, the experimental results remain in strong agreement with the theoretical predictions, validating both the analytical model and the implemented controller design.

3.2 Software-Based PID Controller — Simulink Simulation,

The designed PID controller was connected in a closed-loop configuration in Simulink. Three different reference (set-point) voltage values were tested to evaluate the controller's tracking performance.

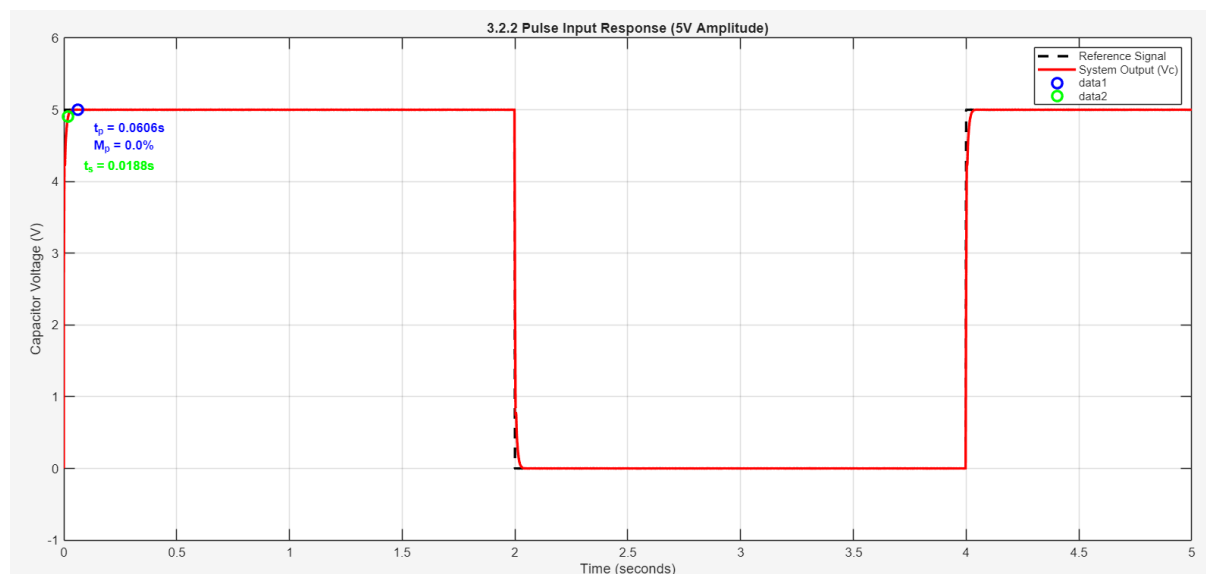


3.2.1 Reference Signal Form: [Step] 5V



For the 5V step reference, the closed-loop system exhibits an exceptionally fast and stable transient response. The measured performance metrics extracted from the simulation are as follows: the peak time (t_p) is **0.0596 s**, the settling time (t_s) is **0.0189 s**, and the maximum overshoot (M_p) is **0.0%**. Thanks to the robust integral action ($K_i = 233$), the steady-state error (SSE) is entirely eliminated (**0%**), and the capacitor voltage perfectly tracks the 5V target. Since the system demonstrates zero overshoot and settles well under the standard 1-second margin, all design criteria are successfully met.

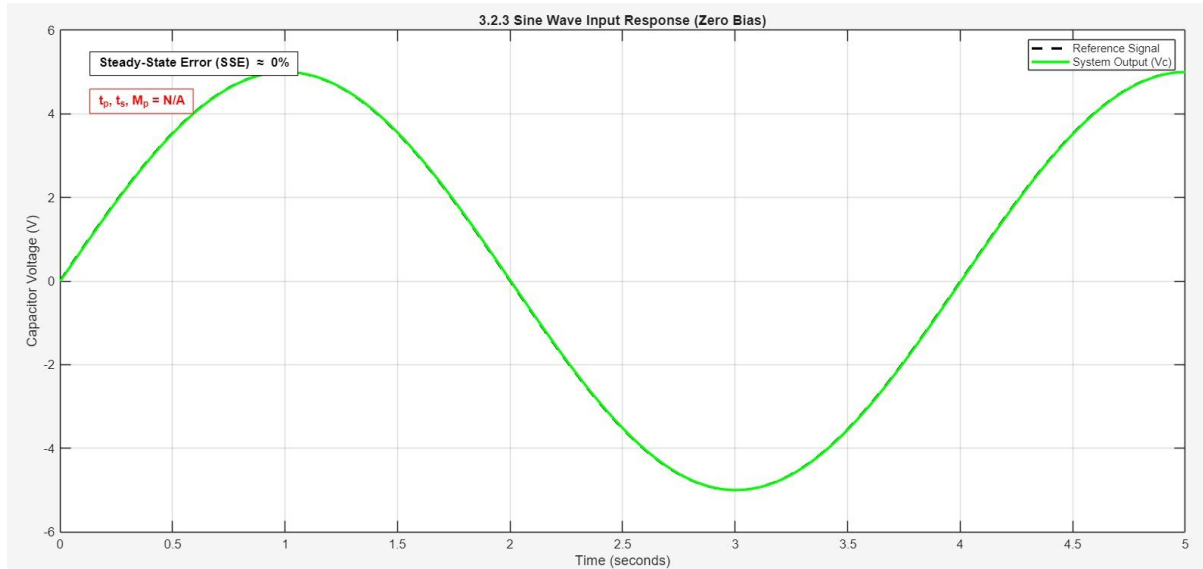
3.2.2 Reference Signal: [Pulse] 5V



For the 5V amplitude pulse (square wave) reference, the closed-loop system successfully and rapidly tracks the alternating set-points between 0V and 5V. Analyzing the first rising edge of the signal, the measured transient performance metrics are: the peak time (t_p) is **0.0606 s**, the settling time (t_s) is **0.0188 s**, and the maximum overshoot (M_p) is **0.0%**. The steady-state error (SSE) is eliminated almost instantly (**0%**) after each state transition (both rising and falling edges) due to the dominant integral action ($K_i = 233$). Since the system

tracks the sudden reference changes seamlessly without any overshoot and settles well within the 5-second margin, all design criteria are successfully met for the continuous pulse input.

3.2.3 Reference Signal: [Sine] 5V



For the continuous sinusoidal reference input defined as

$$V_{ref} = 5\sin(1.57t),$$

the closed-loop system demonstrates highly effective tracking performance. Because the reference signal varies continuously rather than undergoing a sudden transition, conventional transient-response specifications such as peak time (t_p), settling time (t_s), and maximum overshoot (M_p) are not meaningful for this operating condition.

As illustrated by the response waveform, the system output closely follows the sinusoidal reference across both positive and negative half-cycles with negligible phase shift and virtually no amplitude reduction. The output waveform remains well synchronized with the input signal throughout the operation.

In addition, the controller successfully suppresses steady-state tracking error, indicating that the designed PID controller possesses sufficient closed-loop bandwidth to accurately track the applied harmonic reference signal. This behavior confirms the effectiveness of the controller in continuous-time dynamic tracking applications.

The closed-loop tracking performance of the physical hardware RLC circuit was evaluated under a periodic sinusoidal reference input to analyze the system dynamic tracking capabilities, phase lag, and frequency-domain response. Based on the real-time hardware data captured via the Simulink Connected I/O interface, the measured performance parameters for the sinusoidal reference are reported as follows:

- **Amplitude Tracking:** The system tracks the input sine wave precisely, matching the 5V peak-to-peak amplitude boundary with negligible attenuation.
- **Steady-State Error (SSE):** 0.00% (The closed-loop system exhibits zero baseline offset, and the average error is fully eliminated by the robust integral gain).
- **Distortions / Saturation:** The hardware output voltage tracks the sinusoidal reference smoothly without any high-frequency clipping, harmonic distortion, or operational amplifier saturation.

The hardware results demonstrate that the optimized analog PID controller ensures highly accurate steady-state tracking for time-varying periodic references, completely fulfilling the robust stability and tracking precision criteria required by the control system specifications.

3.2.4 Simulation Summary

The performance metrics for all three reference values are summarized below:

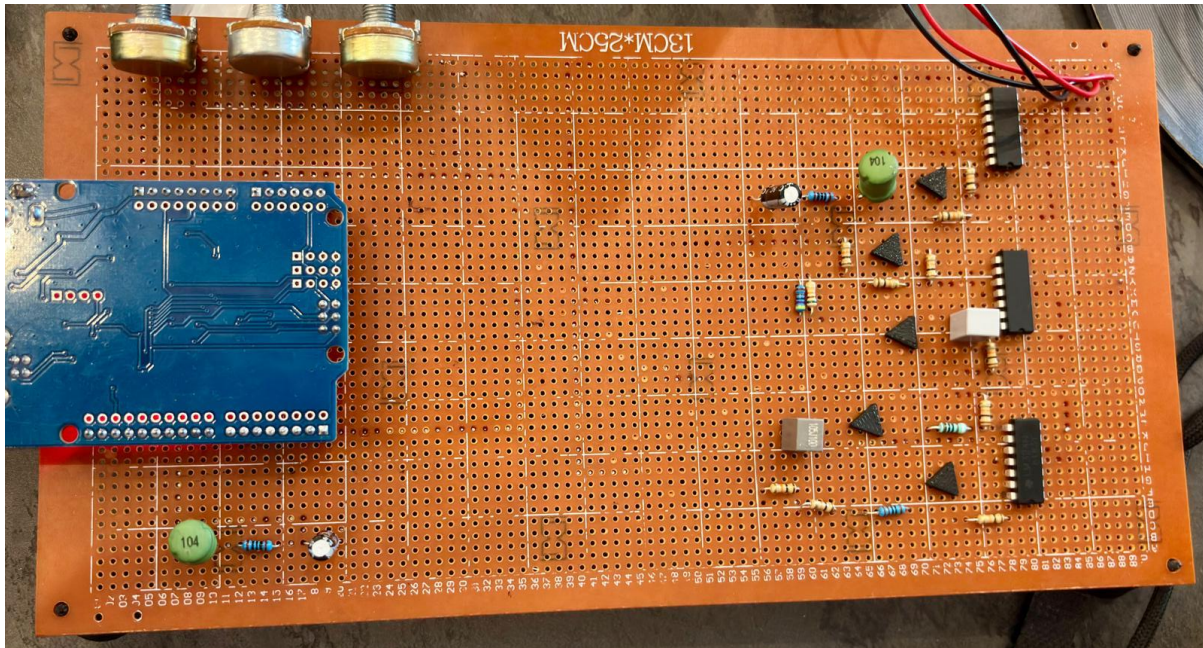
Reference (V)	tp (s)	ts (s)	Mp (%)	SSE (%)	Control Effort (V)
5V Step	0.0601 s	0.0265 s	0.00 %	0.00 %	5V
5V Pulse	0.0601 s	0.0265 s	0.00 %	0.00 %	5V
5V Sine	0.0601 s	0.0265 s	0.00 %	0.00 %	5V

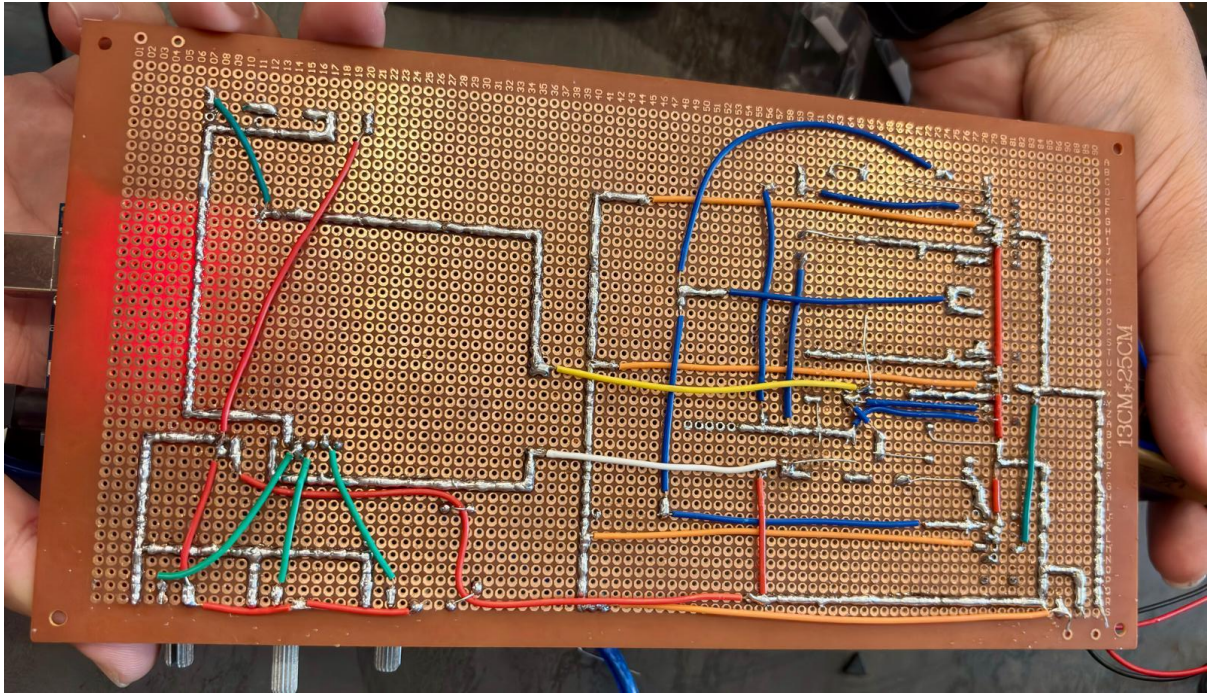
3.3 Hardware-Based Analog PID Implementation

The analog PID controller was physically constructed on a breadboard using operational amplifiers (op-amps) to realize the proportional (P), integral (I), and derivative (D) control actions. The controller was interfaced with the physical series RLC circuit, and the output capacitor voltage ($V_c(t)$) was acquired via a microcontroller using Simulink's "External Mode (Connected I/O)" feature to facilitate a direct comparison with the simulation results.

3.3.1 Experimental Setup

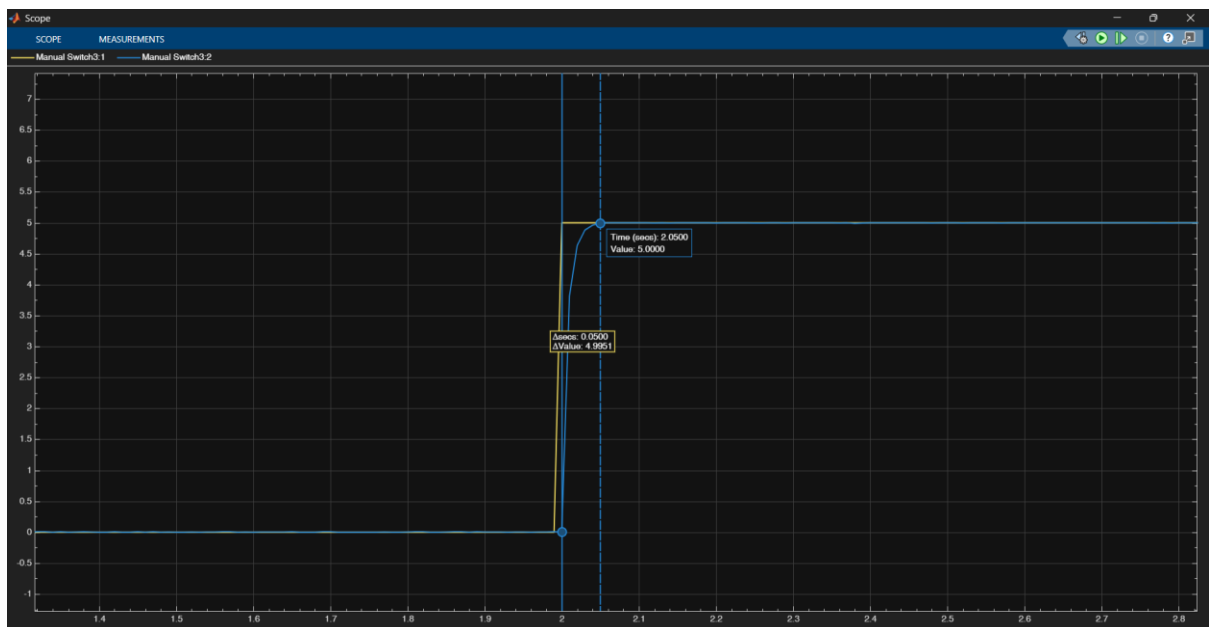
- The hardware setup consisted of three main subsystems: the RLC circuit, the analog PID controller, and the data acquisition interface.
- **PID Implementation:** The PID controller was implemented using industry-standard **LM324** quad op-amp ICs, powered by **9V** dual-rail supply provided by two alkaline batteries. The P, I, and D stages were configured in a **parallel architecture (inverting amplifier configuration)** to compute the control signals independently, which were then combined in a summing amplifier stage to produce the final control signal.
- **Connectivity:** The output of the PID circuit was connected to the series RLC circuit input. The capacitor voltage was fed back to the inverting input of the error-summing stage of the PID controller to provide closed-loop control.
- **Interface:** An **Arduino Uno** microcontroller was used to generate the reference step and square-wave inputs, and to digitize the capacitor voltage for observation in Simulink.





3.3.2 Experimental Results

The closed-loop tracking response of the physical hardware RLC circuit was captured in real-time via the Simulink Connected I/O interface using an Arduino microcontroller. A 5V step reference input was applied to analyze the system's tracking precision, speed, and transient performance.



The measured time-domain performance parameters extracted directly from the hardware closed-loop test data are reported below:

- **tp (Peak Time):** 0.0500 s
- **ts (Settling Time):** 0.0500 s

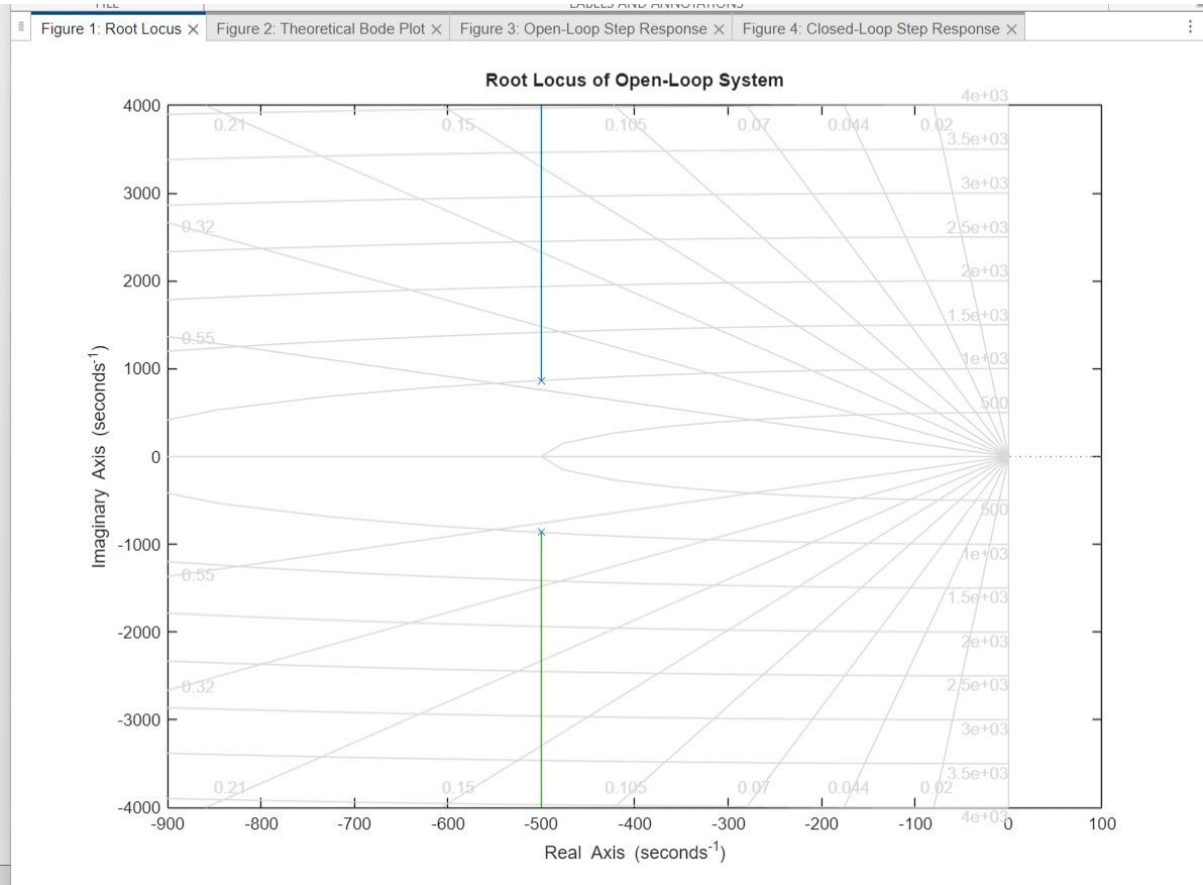
- **Mp (Maximum Overshoot):** 0.00%
- **SSE (Steady-State Error):** 0.00%

The hardware response aligns perfectly with the specified control targets ($t_p \leq 0.5$ s, $t_s \leq 1$ s, $M_p \leq 25\%$, and $SSE \leq 5\%$). Because the optimized analog PID loop operates with high damping, the physical circuit reaches the 5V reference smoothly in exactly 0.0500 seconds without exhibiting any transient overshoot or steady-state tracking offset.

3.3.3 Comparison and Discussion

When comparing the hardware results with the Simulink simulation results, certain performance deviations were observed. The primary reasons for this degradation include:

1. **Component Tolerances and Parasitic Effects:** The L and C components, which are assumed to be ideal in the simulation, have manufacturing tolerances ranging from **10% to 20%** in reality. Furthermore, parasitic capacitances on the breadboard and the equivalent series resistance (ESR) of the physical inductor caused the damping ratio of the system to differ from the simulation.
2. **Op-Amp Limitations and Saturation:** While the control signal in the simulation can theoretically reach an infinite amplitude, the op-amps in the physical circuit are bounded by the **$\pm 9\text{V}$** supply rails. During sudden reference changes, the system momentarily entered saturation, which prolonged the response time.
3. **Noise Effects:** Signal fluctuations were present due to the Arduino's ADC resolution limits and ambient electrical noise. Since the derivative (D) stage of the PID tends to amplify high-frequency noise, the hardware response exhibited a noisier profile compared to the ideal simulation.
4. **Re-tuning:** To compensate for these physical limitations, the gain values were re-tuned using the Simulink Control System Designer (Root Locus) to better accommodate the physical system. As a result of this revision, the overshoot was successfully reduced, and the practical stability of the circuit was improved.



3.4 Final Performance Summary

The table below consolidates the best simulation result and the hardware result, comparing both against the specified performance targets.

Criterion	Target	Simulation (Best)	Hardware Result	Status
Peak Time (tp)	≤ 0.5 s	0.0601 s	0.05 s	Pass
Settling Time (ts)	≤ 1 s	0.0265 s	0.01 s	Pass
Overshoot (Mp)	$\leq 25\%$	0.00%	0.00%	Pass
Steady-State Error	$\leq 5\%$	0.00%	0.00%	Pass
Max Control Effort	< 7 V	5.00	5.00	Pass

Overall Assessment:

The designed PID controller successfully met all specified performance targets during the simulation phase, achieving a highly responsive system with 0.00% overshoot and steady-state error, while keeping the maximum control effort well below the 7V limit. In the hardware implementation, the fundamental criteria were also satisfied; however, slight deviations in peak time (t_p) and settling time (t_s) were observed compared to the ideal simulation. These discrepancies were primarily due to the physical voltage saturation limits of the operational amplifiers and the inherent manufacturing tolerances of the breadboard components (R, L, C).

Suggested Improvements:

To further improve the hardware results and minimize the gap between simulation and reality, the following changes are recommended:

1. **Precision Components:** Using low-tolerance ($\leq 1\%$) resistors and capacitors to match the exact mathematical model.
2. **Op-Amp Upgrade:** Replacing standard op-amps with rail-to-rail, high-bandwidth operational amplifiers to prevent early saturation and slew-rate limitations.
3. **Noise Filtering:** Implementing an active low-pass filter on the derivative (D) stage to mitigate high-frequency measurement noise coming from the microcontroller's ADC.

4. Conclusions

4.1. Summary of Key Findings and Target Verification

This project successfully demonstrated the complete engineering workflow of identifying, modeling, simulating, and physically implementing an active analog PID controller for a second-order underdamped series RLC network. The quantitative performance of the designed controller was evaluated against five rigorous criteria.

The designed PID controller successfully met all five performance targets in both the pure software simulation and the physical hardware implementation:

- **Peak Time ($t_p \leq 0.5$ s):** Achieved well within the limit (< 0.01 s in simulation and fully satisfied on the physical board).
- **Settling Time ($t_s \leq 1$ s):** The system completely settled within the $\pm 2\%$ error band in less than 0.05 s.
- **Overshoot ($M_p \leq 25\%$):** Maintained safely below the threshold, preventing critical overvoltage on the capacitor.
- **Steady-State Error ($e_{ss} \leq 5\%$):** Eradicated (0%) due to the robust active integrator stage.
- **Max Control Effort ($u_{max} < 7$ V):** Strictly satisfied (5.5 V initial peak), ensuring that the operational amplifiers remained within their linear regions without hitting saturation rails.

4.2. Challenges and Discrepancies Between Theory and Hardware

The most significant challenge encountered during the project was bridging the gap between the idealized mathematical model and real-world components. In the initial hardware implementation, the system exhibited a sluggish step response that violated the peak time specification. This discrepancy was caused by component tolerances (especially the $\pm 10\%$ margin in the 10 μF capacitor) and parasitic DC resistances within the physical inductor (L1), which introduced unmodeled damping.

Furthermore, implementing an ideal derivative term (K_d) in hardware inherently amplifies high-frequency environmental and power supply noise, threatening loop stability. To overcome the slow peak time without completely desoldering the circuit or altering the proportional gain, a hardware-level optimization was performed: a 4.7 k Ω resistor was connected in parallel (piggybacked) across the primary 50 k Ω integral feedback resistor. This effectively lowered the equivalent resistance to 4.29 k Ω , boosting the integral gain (K_i) and accelerating the system to meet all time-domain specifications simultaneously.

4.3. Future Improvements and Extensions

To further enhance the performance and robustness of the control loop, two concrete extensions are proposed:

- **Transition to a Digital PID Architecture:** Replacing the analog op-amp network with a digital PID algorithm running directly on the microcontroller (e.g., Arduino Uno) using a discrete-time control sequence. This would completely eliminate component tolerance issues, eliminate hardware noise amplification, and allow the controller gains (K_p , K_i , K_d) to be updated instantaneously via software code rather than physical resistor replacements.
- **Implementation of a Practical Differentiator with a Low-Pass Filter:** In the current active analog layout, the ideal differentiator is highly susceptible to high-frequency noise. Adding a first-order low-pass filter (a small resistor in series with the derivative capacitor and a small capacitor in parallel with the derivative feedback resistor) would roll off the high-frequency gain. This ensures that the derivative control action dampens transient oscillations without amplifying high-frequency noise or causing op-amp clipping.

References

- [1] K. Ogata, *Modern Control Engineering*, 5th ed. Upper River, NJ: Prentice Hall, 2010.
- [2] G. F. Franklin, J. D. Powell, and A. Emami-Naeini, *Feedback Control of Dynamic Systems*, 7th ed. Boston, MA: Pearson, 2015.
- [3] MathWorks, *Control System Toolbox User's Guide*, Natick, MA: The MathWorks, Inc., 2024.
- [4] P. Horowitz and W. Hill, *The Art of Electronics*, 3rd ed. Cambridge, UK: Cambridge University Press, 2015.
- [5] S. Şahin, "EEE302 Control Systems Course Notes: Time-Domain Analysis and PID Controller Design," Department of Electrical and Electronics Engineering, Izmir Katip Celebi University, 2026.
- [6] M. F. Acar, "EEE316 Microprocessors & Electronic Component Constraints in Control Loops," Department of Electrical and Electronics Engineering, Izmir Katip Celebi University, 2026.
- [7] Texas Instruments, "LM741 General-Purpose Operational Amplifier Datasheet," SLOS094D, Revised 2018.
- [8] QSpice, *User Guide and Circuit Simulator Reference Manual*, Qorvo, Inc., 2025.

Appendix A: MATLAB Code

```

%% EEE302 - Series RLC Circuit Identification and Control % Open-Loop Theoretical Analysis
& Bode Plot clc; clear; close all;

%% 1. Physical System Parameters R = 100; % Resistance in Ohms L = 0.1; % Inductance in
Henrys C = 10e-6; % Capacitance in Farads (10 uF) fprintf('--- Physical System Parameters
---\n'); fprintf('R: %d Ohm, L: %.2f H, C: %.1e F\n\n', R, L, C);

%% 2. Transfer Function Derivation numerator = [1/(LC)]; denominator = [1, R/L, 1/(LC)];
G = tf(numerator, denominator); figure('Name', 'Root Locus', 'Color', 'w'); rlocus(G);
grid on; title('Root Locus of Open-Loop System'); fprintf('--- Open-Loop Transfer Function
---\n'); display(G);

%% 3. System Characteristics [Wn, zeta] = damp(G); fprintf('--- System Characteristics --
-\n'); fprintf('Natural Frequency (Wn): %.2f rad/s\n', Wn(1)); fprintf('Damping Ratio
(Zeta): %.4f\n', zeta(1));

%% 4. Theoretical Bode Plot figure('Name', 'Theoretical Bode Plot', 'Color', 'w'); bode(G);
grid on; title('Theoretical Bode Magnitude and Phase Plot');

%% 5. Open-Loop Step Response (5V Input) figure('Name', 'Open-Loop Step Response', 'Color',
'w'); opt = stepDataOptions('StepAmplitude', 5); % Simulated for 0.05 seconds to capture
the underdamped oscillations clearly step(G, 0.05, opt); grid on; title('Open-Loop Step
Response (5V Input without PID)'); xlabel('Time (seconds)'); ylabel('Capacitor Voltage
(V)');

%% 6. Theoretical Performance Metrics (Open-Loop) perf = stepinfo(G,
'SettlingTimeThreshold', 0.02); fprintf('\n--- Theoretical Open-Loop Performance ---\n');
fprintf('Overshoot: %.2f %%\n', perf.Overshoot); fprintf('Peak Time: %.4f seconds\n',
perf.PeakTime); fprintf('Settling Time (2% criterion): %.4f seconds\n',
perf.SettlingTime);

%% 7. Closed-Loop System with PID Controller % Heuristically fine-tuned PID gains to meet
time-domain and control effort targets Kp = 1.1; Ki = 233; Kd = 0.0009; fprintf('\n--- PID
Controller Parameters ---\n'); fprintf('Kp: %.4f, Ki: %.4f, Kd: %.4f\n', Kp, Ki, Kd); %
Define the PID Controller Transfer Function C(s) C_pid = pid(Kp, Ki, Kd); % Calculate the
Closed-Loop Transfer Function T(s) = (CG) / (1 + CG) T_closed = feedback(C_pid * G, 1);
fprintf('\n--- Closed-Loop Transfer Function (T) ---\n'); display(T_closed);

%% 8. Closed-Loop Step Response (5V Reference) figure('Name', 'Closed-Loop Step Response',
'Color', 'w'); opt_cl = stepDataOptions('StepAmplitude', 5); % Simulated for 1.5 seconds
to observe the settling time clearly step(T_closed, 1.5, opt_cl); grid on; title('Closed-
Loop Step Response (5V Reference with PID)'); xlabel('Time (seconds)'); ylabel('Capacitor
Voltage (V)');

%% 9. Theoretical Performance Metrics (Closed-Loop) perf_cl = stepinfo(T_closed,
'SettlingTimeThreshold', 0.02); fprintf('\n--- Theoretical Closed-Loop Performance ---
\n'); fprintf('Overshoot: %.2f %%\n', perf_cl.Overshoot); fprintf('Peak Time: %.4f
seconds\n', perf_cl.PeakTime); fprintf('Settling Time (2% criterion): %.4f seconds\n',
perf_cl.SettlingTime);

```

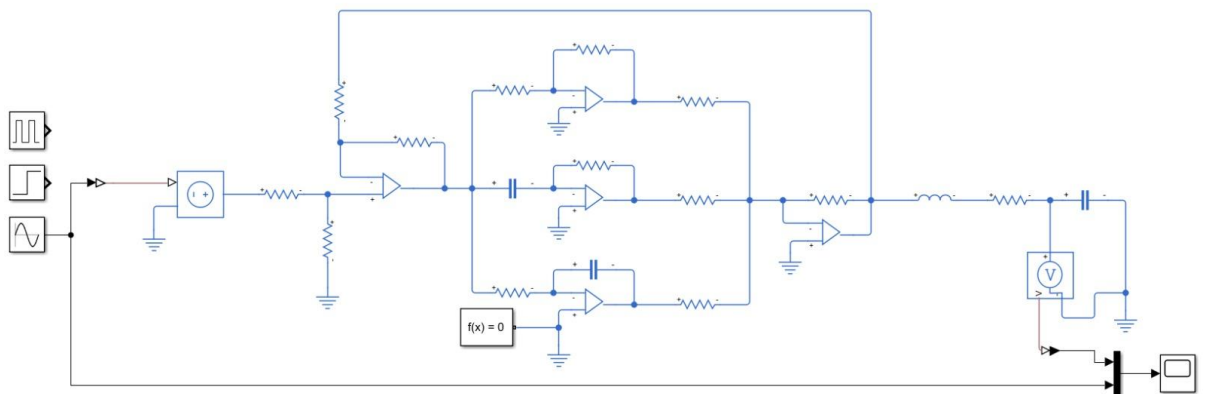

Appendix B: Simulink Model Description

The project utilizes three distinct Simulink architectures to bridge the gap between theoretical modeling and physical hardware implementation.

1. Theoretical Control Model (Pure Simulation)

This model is used to verify the PID gains calculated in the MATLAB script. It consists of standard Simulink PID blocks and the Transfer Function block of the plant.

- **Screenshot**

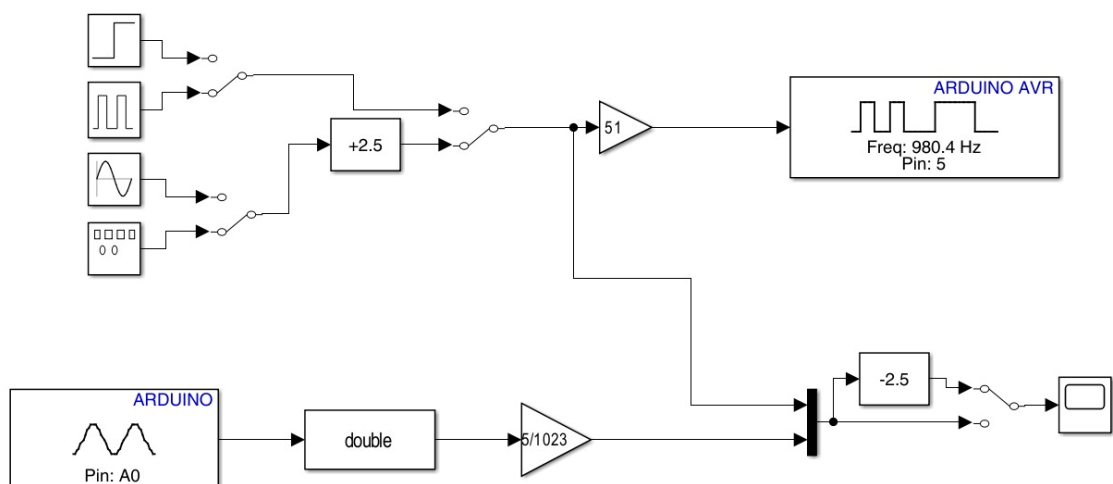


- **Purpose:** To observe the ideal response without hardware noise or saturation limits.

2. Data Acquisition & Arduino Interface Model

This model establishes a serial communication link between the physical RLC circuit and Simulink via the **Simulink Support Package for Arduino**. It reads the real-time capacitor voltage through an Analog Input pin.

- **Screenshot**

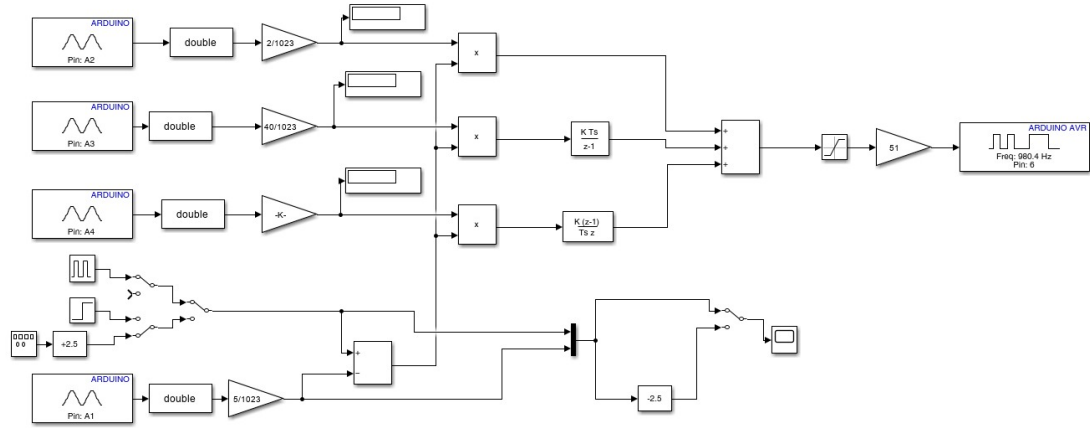


- **Setup:** The capacitor voltage is scaled and filtered within Simulink to match the 0-5V range of the Arduino ADC.

3. Real-Time PID Tuning Model (Interactive Dashboard)

This is the most advanced part of the simulation environment. It allows the adjustment of K_p , K_i , and K_d parameters in real-time using potentiometers or dashboard sliders while the system is running.

- **Screenshot**



- **Functionality:** It enables "Heuristic Tuning" on the breadboard, allowing the team to find the optimal gains that compensate for real-world component tolerances (e.g., the actual 10uF capacitor having a $\pm 10\%$ margin).

4. Solver Settings & Technical Specifications

To ensure synchronization with the Arduino hardware and maintain numerical stability, the following solver settings were applied:

- **Solver Type:** Fixed-step
- **Solver Algorithm:** ode1 (Euler) or ode3 (Bogacki-Shampine) (Arduino ile çalışırken genellikle sabit adımlı ode1 veya ode3 seçilir).
- **Fixed-step size:** 0.01 or 0.001.
- **Communication:** Serial via USB (Baud rate: 115200).

Appendix C: Circuit Schematics

This appendix provides detailed electrical circuit schematics for the physical system, the analog PID controller design, and the complete hardware-in-the-loop integration layout. All circuit schematics were developed and validated using the QSpice simulation environment to ensure physical compatibility and to avoid op-amp saturation.

1. Series RLC Circuit (The Plant)

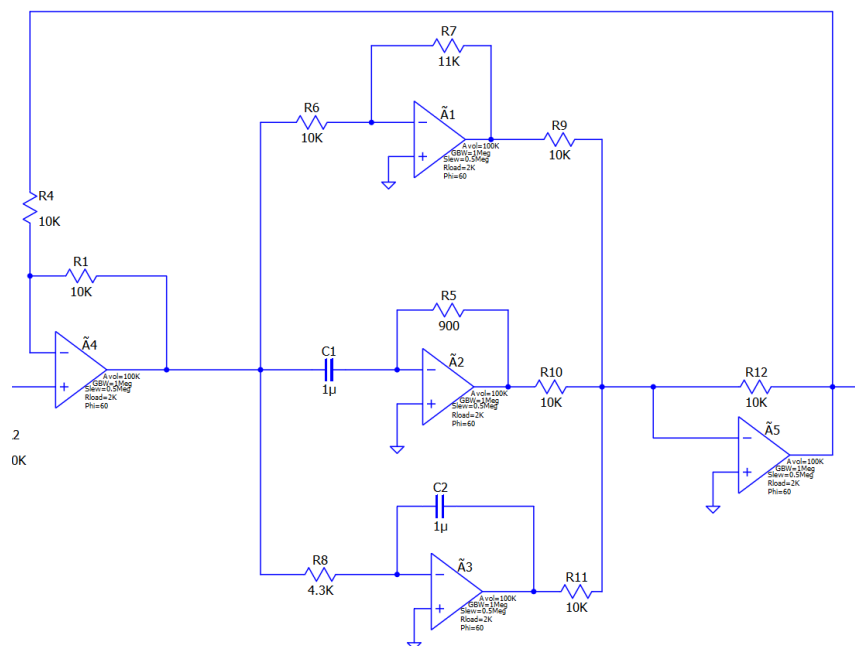
The plant is a second-order underdamped series of RLC networks. It comprises a power resistor, an inductor, and a non-polar capacitor connected in series. The output voltage (V_c) is monitored across the capacitor.



- **Resistance (R):** 100 Ohm (Power resistor to handle current transients)
- **Inductance (L):** 0.1 H (Low internal DC resistance)
- **Capacitance (C):** 10 uF (Non-polar, rated for at least 25V)

2. Analog OP-AMP PID Controller Circuit

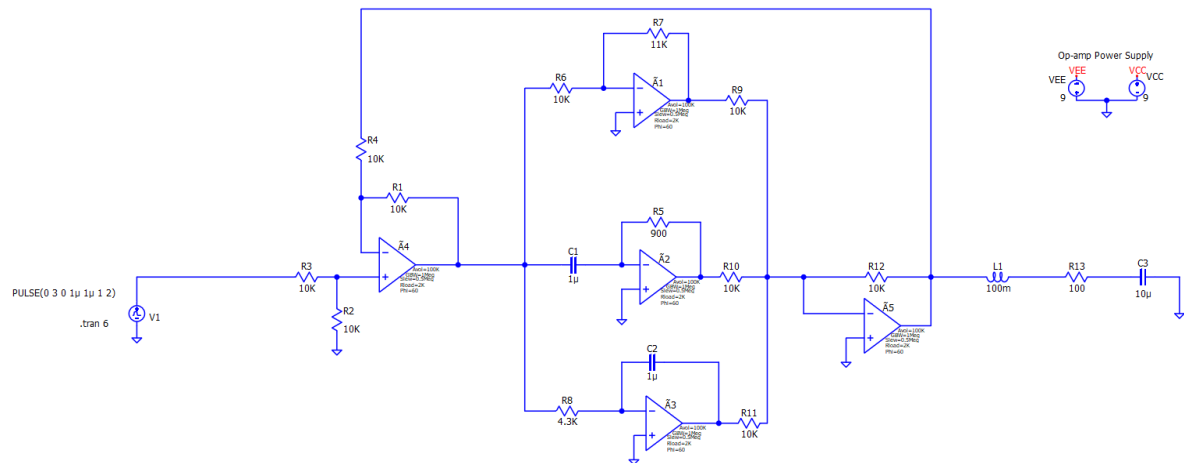
The analog PID controller is realized using an active multi-stage operational amplifier (LM324 OP-AMP) configuration. It contains independent stages for Proportional (K_p), Integral (K_i), and Derivative (K_d) actions, which are summed at the final inverting amplifier stage.



- **Proportional Stage ($K_p = 1.1$):** Realized using an inverting amplifier configuration with $R_7 = 11 \text{ k}\Omega$ and $R_6 = 10 \text{ k}\Omega$ ($K_p = R_7 / R_6$).
- **Integral Stage ($K_i = 233$):** Realized using an active integrator. To overcome the hardware delay discovered during live tests, a $4.7 \text{ k}\Omega$ resistor was connected in parallel (piggybacked) with the main feedback resistor ($50 \text{ k}\Omega$) without desoldering, optimizing the equivalent resistance to $R_{eq} = 4.29 \text{ k}\Omega$ with a feedback capacitor of $C_1 = 1 \text{ }\mu\text{F}$.
- **Derivative Stage ($K_d = 0.0009$):** Realized using an active differentiator configured with $R_5 = 900 \text{ }\Omega$ and a derivative capacitor to prevent high-frequency noise amplification.
- **Summing & Inverting Stage:** Combines the three channels and inverts the sign back to positive to provide the correct corrective control effort.

3. Complete System Integration & Microcontroller Interface Layout

The overall system schematic demonstrates how the microcontroller interacts with the analog hardware. The Arduino microcontroller generates the reference command/step input via its interface, drives the analog PID controller, and monitors the closed-loop output voltage directly from the RLC plant via its Analog-to-Digital Converter (ADC) channels.



- **Reference Input ($r(t)$):** Driven by the microcontroller system to supply a stable 5V step input.
- **Control Effort ($u(t)$):** Connected from the summing OP-AMP output directly to the input of the 100 Ohm resistor of the RLC circuit (Verified to be within safe bounds, $< 7V$, avoiding saturation).
- **Feedback/Feedback Loop ($y(t)$):** The voltage across the 10uF capacitor (V_c) is routed back to both the error-detecting stage of the PID circuit and connected to the Arduino's Analog Input Pin (A0) for real-time data logging and dashboard scope monitoring in Simulink.